Faithful attribution recovers the causal wiring, not the semantics: a ground-truth audit of interpretability on a fully known machine

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Abstract

Interpretability methods are judged by their faithfulness, by whether they name the true causes of a system's output. Yet on real neural systems there is no ground truth against which to check that judgement. We remove the excuse on a system whose ground truth is computable: a bit-exact, differentiable port of the Atari 2600, never built to be interpretable, in which the cause of any output follows from exact intervention. Scoring about 31 methods against this oracle, we find causal methods faithful while popular gradient and correlational saliency methods are provably zero on discrete position outputs (activation patching **1.000** versus vanilla saliency **0.000**), yet score higher on human plausibility. Faithfulness is the smaller half of understanding. An exact method returns a causal-effect map or a data-flow graph, not the computation: a graph-correct circuit need not reproduce behaviour (sufficiency **0.44**), and a feature matching a named variable can be causally inert (matched fraction **1.0**, faithfulness **0.04**). We name this measured residual the *semantic gap*, and stress that we *measure* it, not *close* it. As such, the machine offers a reusable ground-truth benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification. We believe it will help redirect the field toward recovering meaning, not merely confirming causes.

Keywords: interpretability, explainable AI, mechanistic interpretability, faithfulness, ground truth, causal attribution, semantic gap, Atari VCS

1 Introduction

Interpretability now carries real weight, because neural networks drive cars, read scans, and file loans, and before we trust such a decision we ask why. Yet on the systems we most want to explain we cannot check whether the explanation is correct: saliency maps for an Atari agent look convincing and shift with the action, but a convincing map is not a verified one [1]. Hence the field has converged on *faithfulness*, whether an explanation names the system's *true* causes, as its quality criterion. Faithfulness is the right criterion; the difficulty is that on a real network there is nothing to measure it against.

Jonas and Kording made this vivid by turning the neuroscientist's toolkit on a microprocessor whose every transistor is known [2]. Lesions, tuning curves, and connectomics produced rich, suggestive structure, and none of it recovered how the chip computes. Without ground truth, a method that finds structure cannot be told apart from one that finds the *right* structure, and the blind spot recurs across explainable AI, where attribution maps are validated against each other or against human intuition rather than the system's mechanism [3]. The community has let faithfulness stand in for understanding without being able to test the proxy. We remove that excuse.

Paper 1 built the missing reference, a bit-exact, differentiable port of the Atari 2600 Video Computer System (VCS) that reproduces a real, hand-authored 1977 machine to the byte and pixel [4, 5]. The VCS is software with a known specification, so the true cause of any output is computable by exact intervention: we clamp a RAM cell, re-run deterministically, and read off the change. This *intervention oracle* returns, for any candidate cause of any output, the true causal effect, measured rather than estimated, over the platform Fig. 1 lays out with its tiers of ground truth. On this machine, for the first time on a real complex artifact, an interpretability claim can be scored right or wrong.

We turn the toolkit on this system and score it. Across the neuroscience, attribution, and mechanistic-interpretability traditions — 31 methods aggregated from 257 per-game records — we place each method on two axes, faithfulness against the oracle and a documented plausibility proxy for how convincing it looks. The result confirms the causal view, in numbers: causal and intervention methods are faithful, while popular gradient and correlational saliency methods are not. The contrast is sharpest on discrete sprite-position outputs, where the naive gradient is provably zero because position is set by strobe timing, not a value it can flow through. Activation patching reaches faithfulness 1.000 while vanilla saliency collapses to 0.000 on all six core games; aggregated over that regime, causal and intervention methods average 0.412 against 0.068 for the gradient and correlational family, a gap of 0.344. The popular method nonetheless carries the higher plausibility (0.9 versus 0.5), looking most convincing precisely where it is least correct: the field's danger zone of conviction without correctness, which Fig. 4 marks as a measured region.

Securing faithfulness, however, is the smaller half of the problem. With the same oracle we ask not only *which* methods are faithful but *what a faithful method delivers*, and the answer is a true causal-effect map or data-flow graph, not the variable's meaning and not the routine's algorithm. To make this precise we require that an explanation be right only if it is **F**aithful (its claimed causes are the true ones), **S**ufficient (it regenerates behavior under held-out interventions), and **M**inimal at the algorithmic level (right = $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$, Section 3.4). The committed data then show the gap in both directions. An automatically discovered circuit for Breakout is graph-correct and minimal yet does not reproduce the behavior it explains ($\mathbf{F} = 1.0$, $\mathbf{M} = 1.0$, $\mathbf{S} = 0.44$): a correct wiring diagram is not the computed function. A learned feature dictionary matches a named game variable perfectly (matched fraction 1.0) while being causally almost inert ($\mathbf{F} = 0.04$): looking like a known variable is not being how the program computes it. And a concept can be linearly decodable yet provably unused, since what is *present* need not be *used*. The residual between a faithful attribution and an understanding of the computation is not philosophy; on this machine it is a number. We name it the *semantic gap*, the contribution we care about most.

A 1977 chip may seem the wrong place to judge methods built for learned networks. Our answer is that the VCS is a *necessary-condition screen*, not a predictor of success: a method that fails here, with perfect ground truth and no learning obscuring the mechanism, has no business being trusted on a network, where those advantages are lost. The same logic strengthens the harder claim, since if even the strongest causal account a fully observable machine permits stops short of meaning, then on a less observable network faithfulness is *a fortiori* not understanding, and the gap here is a floor on the gap there. The VCS still exhibits the properties that make interpretation hard, from aliased RAM cells to race-the-beam timing; what it does not capture we leave to later papers on learned agents.

Because the claim is sharp, we are deliberate about its limits. We do *not* claim that any method recovered the program's meaning, algorithm, or design from scratch. Every semantic label we use is imported from external annotation sets [6, 7] and only *causally verified* on our substrate, never *discovered* by a method under test, so an alignment method that reports perfect accuracy is right by construction, checking a variable we supplied. Probing shows that information is present, not that it is used [8]; our oracle is the strictly stronger test. As such, Paper 2 *defines and measures* the semantic gap, while *closing* it, recovering a system's documentation and design from the system alone, is the companion work, not this paper. We measure the gap; we do not pretend to have crossed it.

The study thus offers a reusable ground-truth benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification, and reframes the field's open problem: the hard part of interpretability is not faithfulness but the semantic gap that remains once faithfulness is secured, and we believe that naming and measuring it is the first step toward closing it.

2 Related work

Interpretability is studied in three communities that rarely cite one another, yet all ask what, inside a complex system, caused this output. The neuroscience tradition records and perturbs a system, through tuning curves, lesions, connectomics, and directed influence. Jonas and Kording [2] turned this battery on a microprocessor running classic games and found that it produces rich, publishable structure while missing how the chip computes. Their study reported what the analyses found, not how far each fell from the mechanism; we supply that number (Sec. 4).

The attribution tradition asks, for a single decision, which inputs mattered. Gradient and saliency maps [9–12], class-activation methods [13, 14], occlusion [15–17], and surrogate or game-theoretic attributions [18, 19] form the toolkit, applied to Atari agents to visualise what a policy attends [1, 20]. Yet a counterfactual analysis of those same maps found them exploratory, not explanatory [3]: the highlighted regions need not be the causes. On a learned policy that doubt can only be argued, the true cause being unknown; we settle it where it is computable (Sec. 5).

Mechanistic interpretability, the youngest tradition, aims past attribution at the circuit, the subgraph that implements a behaviour. Path patching, causal mediation, and scrubbing intervene on internal activations [21–24]; alignment search and interchange interventions test whether a hypothesised variable is causally realised [25, 26]; sparse dictionaries and probes read features off the state [8, 27–30]. On a real artifact these prove the most faithful (Sec. 6). Yet a recovered circuit, the same oracle shows, is a correct data-flow graph, not the computed function.

The agreed criterion across these traditions is faithfulness: claimed causes should be the true causes. Jacovi and Goldberg [31] make it precise, yet its diagnostics stay indirect when ground truth is absent. Adebayo et al. [32] sharpened this into the sanity check: a saliency method should change when weights or labels are randomised, and several popular methods do not. Yet such checks are only negative: with no reference map none can certify a right method, and the field optimises a proxy it cannot directly score. Our oracle supplies that map and converts faithfulness into a measured quantity: on the discrete sprite-position regime `activation_patching` reaches faithfulness 1.000 while the popular `vanilla_saliency` collapses to 0.000 on all six core games [33], and the leaderboard plots all 31 methods against the same ground truth (Fig. 4) [34].

The concepts we score against are external. AtariARI labels object positions, scores, and lives from commented disassemblies as a probing benchmark [6]; OCAtari adds offset corrections so the coordinates align with the render [7]. These labels are imported, not discovered, and we verify them causally by intervention — set the byte, re-render, confirm the object moves to its predicted place on the bit-exact framebuffer. This is the hinge of our honesty contract: every concept-level (T3) score measures alignment to a supplied label, not a meaning found on its own. Note that a concept can be linearly decodable yet play no causal role, the "present $\neq$ used" trap that control tasks were built to catch [8]; our intervention test is stronger than probing, and that separation is one of the three counted in Fig. 5.

Ours is not the first study to score interpretability against a known answer, yet existing benchmarks build the answer into the system. Yang and Kim [35] grade attribution maps against images with known feature importance; Tracr [36] compiles programs into transformers with circuits known by construction; InterpBench [37] trains semi-synthetic transformers to a specified circuit. Our subject is instead a real, hand-authored artifact whose ground truth is independent of it: never designed to be interpretable, the VCS yields its causal structure to an exhaustive intervention oracle on a bit-exact, differentiable port [4], built on the Atari substrate [38] and the reference emulator [5]. More importantly, those benchmarks validate faithfulness alone, because there the circuit is the meaning by fiat; on a real artifact the meaning is a separate object from the causal structure. Hence we can do what none of them can: separate faithfulness from understanding, and show that a method which perfectly recovers the true wiring has not recovered the semantics (Fig. 5). We measure that residual, the semantic gap, and are explicit that closing it — recovering a program's variables and design from scratch — is a different problem, the subject of companion work; we claim no unqualified "first".

Our use of "interpretable" is one of several in circulation. We operationalise it as ground-truth recovery: an explanation is interpretable to the degree that its claimed structure matches the system's true causal structure, scored by the oracle. Barbiero et al. [39] instead define it as inference equivariance, a design target for new models, while ours audits a system already known. Read against Marr's levels of analysis [40] and Lazebnik's "fix-the-radio" parable [41], the point that drives this paper holds: a faithful description of a system's wiring is not yet an account of the algorithm it runs, and on a known machine that gap is measured.

3 Methods

Our method rests on a single idea: we study an interpretability question on a machine whose answer we can compute exactly. We describe in turn the substrate that makes the ground truth obtainable, the oracle that delivers it, the tiered semantics we attach to it, and the correctness triad that turns "do we understand?" into a set of numbers. The per-method setups live in the Supplementary Information; Fig. 1 sketches the pipeline.

3.1 The differentiable, bit-exact VCS substrate

The subject of this study is the Atari Video Computer System (VCS) itself, not a learned agent: a complex real artifact never designed to be interpretable, yet whose every causal dependency we can read off exactly. Paper 1 built it, a bit-exact, differentiable port of the 1977 VCS validated bit-for-bit against the `xitari` C++ reference on all 64 games of the Paper-1 roster, in random-access memory and rendered screen, within the validated conformance window [4, 5]. We use it in two interchangeable, mutually cross-checked realizations, `jutari` (Julia) and `jaxtari` (JAX/Python).

The substrate grants four capabilities that no neural network offers for free. The first is an exact forward map,

$$y \;=\; \mathrm{VCS}\big(\mathrm{ROM},\, \mathrm{inputs}[0{:}t],\, s_0\big), \tag{1}$$

that is, the output y at step t, a pixel, a score, or a game event, is a deterministic function of the cartridge bytes, the inputs up to t, and the initial state s_0, and re-running reproduces y to the bit. The second is exact interventions: we set any cause u, a ROM byte, a RAM cell, a CPU/TIA/RIOT register, an opcode, or a joystick input, to any value and re-run Eq. (1) deterministically, realizing the causal operator $\mathrm{do}(u)$ on a real machine rather than an assumed world model. The third is full state observability, every bit of RAM and every register recorded along the trajectory, so the "activations"

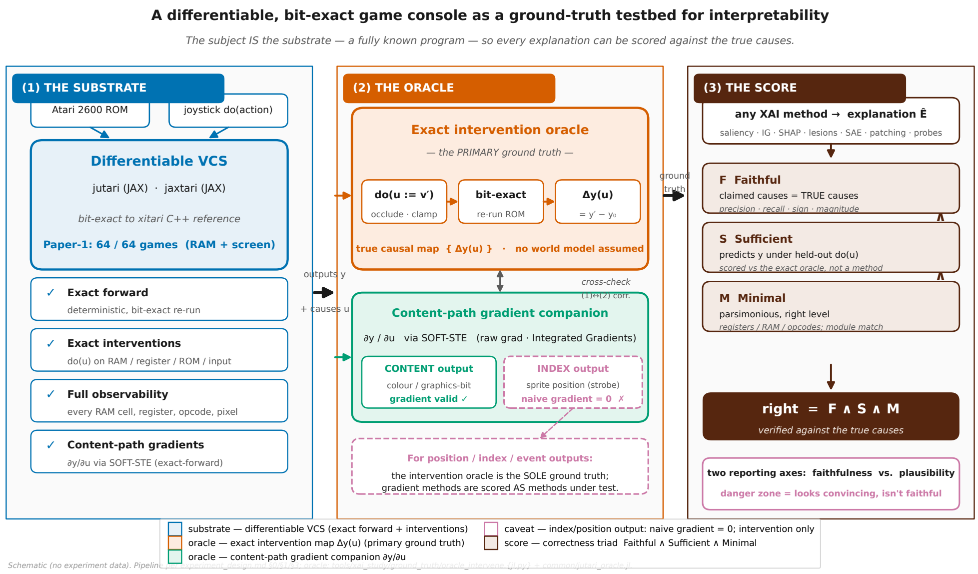

Fig. 1 The measurement platform and its ground-truth oracle. A differentiable, bit-exact port of the Atari VCS (left) gives four capabilities no neural network grants for free — exact forward, exact interventions $\mathrm{do}(u)$, full state observability, and content-path gradients. On it we build the oracle (centre): the primary *exact intervention oracle*, $\mathrm{do}(u{:=}v') \to \Delta y(u)$ by bit-exact re-run, and a *content-path gradient* companion $\partial y/\partial u$. Any explanation $\hat{E}$ is then scored by the correctness triad (right), right = F ∧ S ∧ M, on the two reporting axes (faithfulness vs. human plausibility). *Note that* a discrete index — a sprite's position, set by strobe timing — has zero naive gradient; for such position/index/event outputs the intervention oracle is the *sole* ground truth and gradient methods are scored as methods under test. The schematic draws no experiment data.

we interpret are the program's own variables. The fourth is a content-path gradient $\partial y/\partial u$ through a straight-through estimator (STE) formulation whose forward pass is bit-exact to the hard machine at any finite temperature [42, 43], batched on the GPU. The inputs we drive, the state we read and clamp, and the outputs we explain are the three faces a neural interpretability study works with, so one substrate carries all three batteries.

We honor one boundary throughout. Paper-1 bit-exactness holds within a bounded frame window on a fixed action stream, on the order of 30 frames of RAM and 60 frames of screen, and every scored claim is taken inside that validated horizon. Long-horizon end-to-end gradients are not needed for our single-step metrics, so we do not assume them.

3.2 The ground-truth oracle

The oracle is the heart of the method: for any output y and any candidate cause u it returns the true effect of u on y, against which every method is then scored. Its primary construction is the exact intervention oracle: we intervene on u by occlusion, clamping, replacement, or resampling, re-run the machine deterministically, and record the change in the output,

$$\Delta y(u) \;=\; \mathrm{VCS}\big(\ldots,\, \mathrm{do}(u{:=}v'),\, \ldots\big) \;-\; y. \tag{2}$$

The true causal effect of a cause is thus the bit-exact difference its surgical edit makes to the output, measured by actually running the altered machine, with no world model or smoothness assumed. The collection $\{\Delta y(u)\}$ over all causes is the true causal map of y, the reference for the faithfulness axis of every method.

The content-path gradient $\partial y/\partial u$, read through the differentiable substrate, is its companion construction. It is fast and dense, and it agrees with the intervention oracle on the content path, the register, color, and graphics-bit values that flow smoothly into the pixel. Yet, a single decisive caveat carries through the whole study. The STE gradient routes to the content path only, so a discrete index, a sprite's screen position the VCS sets by the timing of a strobe write, has zero naive gradient, because an infinitesimal change of the cause never crosses the integer boundary that moves the sprite, and recovering it requires the bilinear sampler of Paper 1 [44]. Hence, for content outputs the gradient is a legitimate companion oracle, while for position, index, and event outputs the intervention oracle of Eq. (2) is the sole ground truth and every gradient-based method is scored as a method under test. This position regime is where the sharpest separations in our results appear. Where the two overlap we report their correlation and flag, rather than hide, the points at which they disagree.

3.3 Tiered ground truth and how we obtain it

The ground truth comes in three tiers, and the tiering is not bookkeeping; it carries the central argument of the paper. Tier 1 (T1) is the exact causal map of Eq. (2), derived internally from the substrate and available for all 64 games. Tier 2 (T2) is the hardware roles and the read/write data-flow graph, also exact and internal across all 64 games. T1 and T2 are everything the machine *is* and need no external label.

Tier 3 (T3) is different in kind, and that difference is the point. It attaches a game-concept name to a cell or feature, "the ball's x position" or "the score," names not computable from the substrate alone, because they are facts about what the program *means*, not what it *does*. We import T3 from two public sources and make it causal. AtariARI provides labels for 22 games, read from commented disassemblies, with raw RAM coordinates not aligned to the rendered position [6]; OCAtari covers more than 40 games and, in its RAM mode, applies the render-time offsets that fix that misalignment [7]. We import both as candidates and verify each causally by setting the byte, re-rendering, and confirming on the bit-exact framebuffer that the named object moves to the predicted place, which upgrades a correlational label to a verified-causal one and auto-corrects the offsets; further labels we discover by RAM-to-framebuffer correlation and intervention. The headline core set is six games, Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, and Q*bert, each carrying both sources, all in the Paper-1 bit-exact roster, together spanning paddle-and-ball, shooter-with-shields, resource-meter, multi-agent-pursuit, and discrete-grid logic. The remaining games form a breadth set scored on T1 and T2 only.

We are deliberate about what this buys us. All T3 labels are external and only verified, never discovered from scratch, so a method that matches a supplied concept has aligned to a given meaning, not recovered it. This is exactly why a probe can show that information is *present* without showing it is *used* [8], and why our intervention test, which asks whether the byte *causes* the named object to move, is strictly stronger than decoding it. Recovering the program's documentation and design from the bare machine is a separate problem, the subject of the companion paper.

3.4 The $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$ correctness triad

A faithful attribution is not yet an understanding, and to make that a measurement we need an operational criterion. We adopt one in the spirit of Marr's levels of analysis [40] and of Jonas and Kording's warning that finding structure is not the same as explaining it [2], calling an explanation $\hat{E}$ of how an output y arises *right* if and only if it is Faithful, Sufficient, and Minimal,

$$\text{right} \;=\; \mathrm{F} \,\wedge\, \mathrm{S} \,\wedge\, \mathrm{M}, \tag{3}$$

that is, it must name the true causes, regenerate the behavior, and do so at the right level with no spurious parts. The three axes are defined against the oracle, not against any method under test, which keeps the criterion non-circular.

Faithfulness asks whether the claimed causes are the *true* causes. We score $\hat{E}$ against the intervention oracle of Eq. (2) by correlation, sign and magnitude agreement, deletion and insertion area-under-curve, and precision at k against the true causal top-k. It is the field's usual quality criterion, and on our machine it can be checked. Sufficiency asks whether the explanation could regenerate the behavior under interventions it has not seen; we score it against the exact oracle by predicting y from $\hat{E}$ under a held-out $\mathrm{do}(u)$ and comparing to the true y on the bit-exact re-run. We deliberately do not define it through interchange interventions or scrubbing, which are themselves mechanistic-interpretability methods under test in Phase C, since that would make the criterion circular. Minimality asks whether the explanation is parsimonious, free of spurious parts, and lives at the algorithmic level of the registers, RAM cells, opcodes, and program variables of the machine's own module decomposition, matching the computation's known structure, not merely being small.

Two reporting axes carry the headline comparison, faithfulness against the ground truth and human plausibility. Plausibility is a documented method-tradition proxy, not a human-subjects measurement, and we report it as such; its job is to expose the danger zone of explanations with high plausibility but low faithfulness, convincing yet wrong. Separating F, which the wiring satisfies, from S and M, which the *computation* must satisfy, is what lets us name, later, the residual that faithfulness alone leaves behind.

3.5 Methods under test, games, and aggregation

We turn three traditions onto the machine and score each by the triad of Eq. (3). The neuroscience battery (Phase A) runs connectomics and data-flow recovery, single-unit lesions, tuning curves, pairwise and global correlation structure, pooled-activity spectra, Granger causality, dimensionality reduction, and whole-state recording, scoring each against the known mechanism to quantify Kording's lesson at the low-faithfulness baseline [2]. The attribution and XAI battery (Phase B) runs gradient saliency and its variants, integrated gradients, occlusion, perturbation masks, randomized masking, LIME, and Shapley-style attribution [9, 10, 15–19]; Grad-CAM and its variants and attention rollout [13, 14, 45] require neural layers or a learned policy and are recorded as not applicable. The mechanistic-interpretability battery (Phase C) runs activation patching and causal mediation, interchange interventions and distributed alignment search, attribution and path patching, automatic circuit discovery, sparse autoencoders, causal scrubbing, linear probing with control tasks, and the logit and tuned lens [8, 21–30, 46].

All faithfulness scoring runs on the six core games of Section 3.3, where the oracle's re-runs are exact. Cell- and pixel-level scores, correlation, deletion and insertion area-under-curve, and precision at k, are taken against T1 and need no T3; only object- and concept-level scores draw on the external T3 labels, and we flag every such score as alignment-to-given, not discovery. We aggregate the committed per-game records into one row per method on the shared faithfulness-versus-plausibility axes of the Results leaderboard. The headline contrast is the position regime, where the naive gradient is provably zero. There, causal and intervention methods average a faithfulness of 0.41 against 0.07 for gradient and correlational methods, a gap of 0.34, while the per-method extreme places an exact causal method, activation patching, at faithfulness 1.000 and the field's default saliency at 0.000 on all six games, the popular method nonetheless carrying the higher plausibility proxy (0.9 versus 0.5). Against these numbers the semantic gap is measured.

4 The neuroscience battery, quantified

We begin where the field's doubt began. Jonas and Kording asked whether a neuroscientist with the standard toolkit could understand a microprocessor, and found that the toolkit yields rich structure that misses the mechanism [2]. They could diagnose the failure but not grade it, because on the 6502 they had no causal ground truth to score each method against. We supply that number. Running the same battery on the bit-exact, fully known VCS, where the true causal role of every register and RAM cell is recoverable by exact intervention (§3), turns Kording's qualitative lesson into a measured one. Across the eight analyses A1–A8, faithfulness against the oracle ranges from 0.116 to 1.0 and averages 0.49 ± 0.24 [34], while the oracle-as-method positive control sits at 1.0 by construction (Fig. 2). The structure these methods report is real; yet structure is not cause, and the gap is what the score measures.

The wiring itself is largely recoverable. A1 reconstructs the dependency graph over state variables, a connectome of read/write edges, and scores it against the true dataflow graph at a faithfulness of 0.41 [34], half spurious. The single-unit lesion A2 fares best: clamping each RAM cell in turn and ranking it by the behavioral damage it does yields an importance map whose Spearman correlation with the unit's true causal role is 0.99 (mean 0.987 over the six core games) [34]. Lesioning is causal, and the one classical number near the ceiling proves it.

Yet a near-perfect rank order hides an interaction blind-spot. A single-unit lesion changes one cell at a time, so it is blind to causes that act only in concert. On Space Invaders the per-unit ranking misses 25% of the interactions the oracle records, and one super-additive pair swings behavior by up to 1044 screen-pixels beyond the sum of its parts [34]. The map of which units matter is right, the model of how they combine absent. Hence even the strongest classical method recovers a ranking, not the computation: an effect map is not the algorithm that produces it.

Tuning curves report structure that points the wrong way. A3 fits per-unit tuning to a game variable, the workhorse of systems neuroscience, and on the VCS it actively misleads. Its *spurious-tuning rate*, the fraction of strongly-tuned units whose tuning does not match their true causal role, is 1.0 on Pong, and its faithfulness collapses to 0.12, the lowest in the battery [34]. Tuning is correlational; on a machine where many cells co-vary with the beam clock, apparent tuning is cheap, which foreshadows the present-versus-used distinction we return to for linear probing.

Correlation structure and field potentials are real and largely epiphenomenal. A4, the spike-word analysis of pairwise and global correlation, reproduces the familiar weak-pairwise/strong-global signature, yet against the true coupling its faithfulness is only 0.29 with sufficiency near zero ($S = 0.018$) [34]. A5, the local-field-potential analogue pooling regional activity into power spectra, scores 0.40, much of that variance being the known frame and scanline clocks [34]. Both see the machine breathe without seeing it compute.

Granger causality mistakes temporal precedence for cause. A6 infers directed CPU/TIA/RIOT edges and scores them against the true data-flow; on Pong its faithfulness is 0.0, with false-edge and missed-edge rates both at 1.0 [34]. The VCS is clock-locked, nearly everything predictable from everything else one cycle earlier, so equating precedence with causation infers spurious edges everywhere. No longer lag or stricter threshold repairs that; precedence and causation come apart here, and only the intervention oracle tells them apart.

Dimensionality reduction lands in the middle, NMF edging out PCA. A7 decomposes the state tensor into latent components and matches them to the known clock, read/write, and vsync signals; non-negative matrix factorization recovers a matched-component fraction of 0.60 (best 0.80 on a single game), slightly ahead of PCA at 0.60 on average, its non-negativity a better fit to the substrate [34]. Neither clears two-thirds of the known components, so A7's 0.60, second-strongest behind only the causal lesion, leaves the latent space half interpretable and half opaque.

A perfect record is still not an explanation. A8 records the full RAM and register state, faithful and sufficient by definition ($F = 1.0$, $S = 1.0$), and yet its minimality is 0.07 [34]: a recording of everything is, at the algorithmic level, an explanation of nothing. Faithfulness and sufficiency without minimality is a log file, not an account of the computation, the first hint that the three axes pull apart, which the semantic-gap result later turns into the paper's centre of gravity.

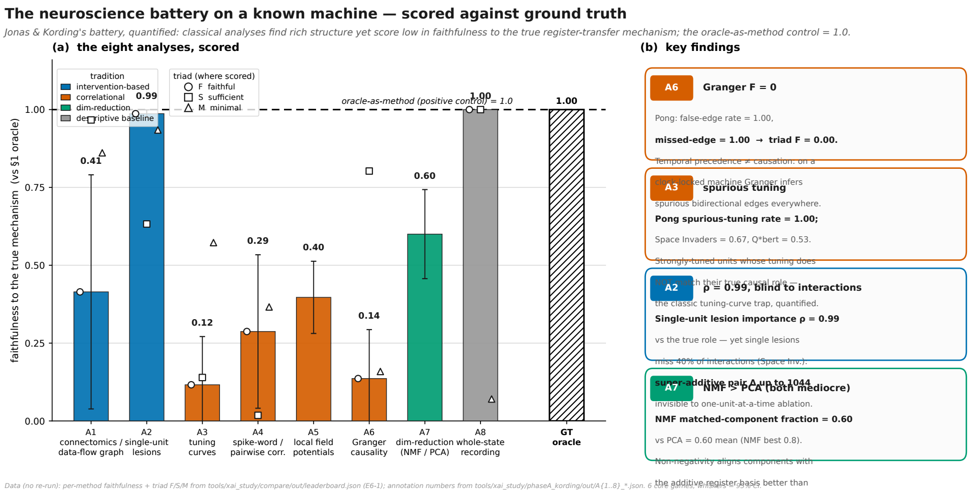

Fig. 2 The Kording neuroscience battery, scored against the exact oracle. Each analysis A1–A8 is run on the bit-exact, fully known VCS and graded for faithfulness against the §3 intervention oracle; bars are coloured by method tradition, and the oracle-as-method positive control is drawn at faithfulness = 1.0 (dashed reference). The classical battery averages 0.49, with only the causal single-unit lesion (A2) and dimensionality reduction (A7) clearly above chance. Note that the four callouts isolate the mechanism behind the low scores: A6 Granger causality reaches $F = 0$ on Pong (temporal precedence read as causation on a clock-locked machine), A3 tuning shows a spurious-tuning rate of 1.0 on Pong (strongly-tuned units whose tuning is not their true causal role), A2 attains a single-unit importance correlation of $\rho = 0.99$ yet misses 25% of interactions with a super-additive pair swinging up to 1044 pixels, and A7 NMF edges out PCA on matched components while both stay below two-thirds. The figure is the calibration baseline that the causal contrast of Fig. 4 sharpens. Every plotted value reads from a committed per-game record; no number is re-run or estimated.

Taken together, the battery confirms the expected result and puts a number on it. The advance over Jonas and Kording is that we grade the *architectural* register-transfer level with a *quantitative* F against an exact oracle, not a qualitative verdict on the transistor. The contrast is sharp, because the causal operators we turn on the same machine in Sections 5–6 reach the ceiling this battery cannot. As such, the gap is the method, not the machine. We mean this strictly as calibration: even the battery's best members recover a ranking, a graph, or a record, never the algorithm the program runs. That residual is the problem the paper measures.

5 Results B — attributing a VCS output to its causes

Attribution hands back a map of which inputs mattered, and we turn the toolkit on the VCS to ask the one question a real network never lets us ask, whether the map is true. Every attribution is scored against the same §1 intervention oracle that defines ground truth (Sec. 3), so for each chosen output y, a pixel, a score, a game event, we know the exact set of causes it should have named. We ran twelve attribution methods across

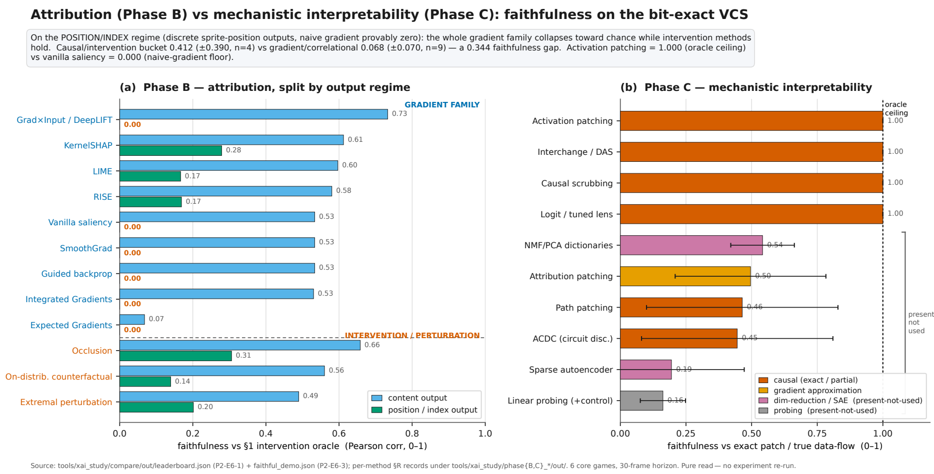

Fig. 3 Attribution (Phase B) and mechanistic interpretability (Phase C) scored against the same §1 intervention oracle. The Phase-B half (a) splits each method's faithfulness by output regime: on a *content* output the gradient family lands on the true cause, but on a *position*/index output the entire gradient family *collapses to* 0.00, while the intervention methods hold up because each probe is a real re-run of the bit-exact program. Note that the popular gradient methods, which fail on position, carry the higher human-plausibility proxy, the danger zone of Sec. 7 in numbers. The Phase-C half (b) is discussed in Sec. 6. Vector figure; values read from the committed cross-tradition leaderboard.

six core games (Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, Q*bert) and an oracle-as-method positive control, placed on the shared faithfulness axis in Figure 3 (Phase-B half) and scored per method in Table 1.

The headline is a split, not a ranking. A content output — the value of a register, a colour, a graphics bit — is a differentiable function of the state, so its gradient flows; a position output — where a sprite is drawn, set by strobe timing on a discrete index — has provably zero naive gradient (Paper 1, and Sec. 3). Hence the entire gradient family inherits this boundary, and it cuts the leaderboard cleanly in two.

On the content path the gradient is faithful and unremarkable. Vanilla saliency [9] lands its mass on the true causal byte: on Q*bert its attribution of a content read correlates with the oracle at Pearson 0.999 with precision@3 of 1.0, on Ms. Pac-Man at 0.811 / 1.0, on Breakout at 0.750 / 1.0. Integrated Gradients [10], Grad×Input/DeepLIFT [47], SmoothGrad [11], and guided backpropagation [12] all reproduce this, because the output we explain is a one-hot read whose Jacobian $\partial y/\partial u$ is a constant pointing at the read index. Smoothing leaves the score put ($\Delta\,\mathrm{corr} \approx 0$) and guided backprop degenerates to vanilla saliency; on this path there is nothing for the methods to disagree about.

The position output is where the family fails, and it fails by vanishing. The naive gradient is identically zero on every game, so saliency, Grad×Input, SmoothGrad, guided backprop, and Integrated Gradients all collapse to a faithfulness of 0.0 in the position regime — the "Fail / zero on index outputs" prediction of Sec. 3, now a measured number. Averaged over all output regimes the gradient family scores only 0.294 faithfulness against the oracle (Fig. 3); on the position regime alone, 0.068. Returning the empty map for half a game's outputs is not a partial success but a structural blind spot, exactly where the discrete game logic lives.

Two finer results sharpen the warning. The baseline choice, the perennial worry about Integrated Gradients, is a magnitude effect here, not a faithfulness one: the $(x - x_0)$ factor moves max $|\mathrm{IG}|$, yet the ranking and the oracle correlation hold, because the constant one-hot Jacobian keeps the mass on the same byte; only the degenerate baseline equal to the content byte breaks it. Expected Gradients [48], averaging IG over a distribution of recorded VCS states, removes that degeneracy and is provably stable across reference draws (mean pairwise cosine ≈ 1). Yet, on the content path it scores the same faithfulness as a well-chosen single-baseline IG at $M\times$ the compute and still vanishes on position; the more careful method buys stability, not truth, and its all-regime faithfulness of 0.034 is the lowest in the study.

Guided backprop and vanilla saliency fail the program-randomization sanity check of Adebayo et al. [32] on the system itself. We scramble the ROM so the program boots to a completely different state ($\sim$95–127 of 128 RAM bytes change), and the content-path saliency map stays byte-identical (cosine ≈ 1.0 on all six games). The Jacobian $\partial y / \partial u = e_{\mathrm{idx}}$ is a constant one-hot, independent of what the program computes, so the map points at the read index no matter which program produced the output, i.e. a saliency that is provably model-invariant. A faithful explanation must change when the model changes; this one cannot, whereas the intervention oracle distinguishes the two programs at once. The map lands on the right cause yet stays blind to the model that produced it.

Replace the gradient with an actual intervention and the position blind spot disappears. Occlusion [15] sets a candidate cause and re-runs the bit-exact program, the coarse twin of the oracle; on the Pong position output, where every gradient method scores 0.0, it recovers the true causal bytes at correlation 0.837, its all-regime faithfulness of 0.482 the highest in Phase B. The black-box perturb-and-re-run methods follow the same logic and result: RISE [17] (0.375), LIME [18] (0.382), and KernelSHAP / Shapley sampling [19, 49] (0.446) each draw masked coalitions and average over the genuine responses, so each works on position outputs where the gradient is dead, and each reports its own completeness ($\Sigma\phi = y(\text{intact}) - y(\text{all-masked})$ for KernelSHAP, the surrogate R^2 for LIME).

The two minimal-mask methods meet Atrey et al.'s [3] objection that perturbations stray off-manifold. Extremal perturbation [16, 50] optimises an area-bounded mask of real *do*(occlude) re-runs, and the on-distribution counterfactual [51] substitutes candidate cells toward a genuine alternative VCS state, the RAM of another frame of the same ROM. Both stay on the data manifold by construction and both work on position outputs (faithfulness 0.346 and 0.350). As such, the intervention family averages 0.681 faithfulness against the oracle, more than double the gradient family's 0.294 (Fig. 3): a perturbation is faithful exactly when it is a valid intervention.

A part of the popular toolkit returns no answer at all, because it has nothing on the VCS to attach to. Grad-CAM and Grad-CAM++ [13, 14] need a final convolutional layer to weight; attention rollout [45] needs attention matrices; VIPER [52] needs a learned policy to distil. The VCS is a 6502 CPU with a TIA and a RIOT executing a fixed ROM, so across all six games the count of each required substrate is exactly zero, against ≥ 16 genuine candidate causes per game for the applicable methods. We record this as a measured finding, `applies` = `false`, not a poor score: the absence is structural and the same in every game, and re-targeting Grad-CAM's pooling onto raw register state would only be the content-path gradient renamed. That a slice of the field's headline methods is silent on a real, deployed artifact is itself part of the audit.

Across the twelve methods the leaderboard separates along the axis that matters and crosses the axis the field actually optimizes. The starkest pair makes it concrete: on the position regime the causal method activation patching (Phase C) reaches faithfulness 1.000, the oracle ceiling, on all six games, while vanilla saliency, the field's default tool, collapses to 0.000 on all six, a per-method gap of 1.000, bit-exact on every game. Yet, on the human-plausibility proxy of Sec. 3 the popular method carries the higher score (0.9 versus 0.5): the provably empty map looks more convincing than the one that is exactly right. That inversion of low faithfulness paired with high plausibility is the danger zone, and on the VCS it is not a worry but a measurement.

We claim no more than the faithfulness of these maps. A correct occlusion map names the true causal bytes; it does not tell us what those bytes mean, that a cell is the ball's x or that a routine is the bounce. Object-level semantics enter only through the external T3 labels (AtariARI, OCAtari; Sec. 3), and even there we verify a supplied label by intervention rather than discover it. Whether a faithful attribution amounts to understanding the computation we take up in Sec. 7; here we have established only that faithfulness can be had, and that the most popular way of seeking it does not deliver it.

6 Mechanistic interpretability on a known circuit

Mechanistic interpretability makes the strongest claim in the field: not where a system attends, but how it computes. Its methods read internal activations as a circuit and try to recover that circuit, and on a neural network the recovered circuit can only be checked against itself, because the true wiring is unknown. We remove that excuse. The VCS's data-flow is the program's data-flow, known exactly from the disassembly, and every intermediate value is a RAM cell or a register we can read, clamp, and re-run bit-exactly. We therefore turn ten mechanistic-interpretability methods onto a circuit whose answer is already on the page, and score each against the same §4 oracle: the exact intervention table, the true read/write graph, and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad of §3. To the best of our knowledge this is the first calibration of the toolkit against a *known* circuit inside a real, hand-authored artifact rather than a synthetic or compiled one. Figure 3b places the Phase-C methods on the shared faithfulness axis. The result splits cleanly, and the split is the point of the paper: the causal methods are exact, yet exactness buys the wiring, not the meaning.

Exact recovery of the wiring

The headline is bare. Activation patching recovers the true causal-effect table exactly: across all six core games the recovered per-site effect equals the exact intervention

Table 1 Phase-B attribution scored against the §1 intervention oracle (six core games; oracle-as-method positive control = 1.0 on every non-degenerate column). *Faith* is the all-regime faithfulness against ground truth and *Faith (pos)* the position/index regime alone, where the naive gradient is provably zero; *Plaus* is the documented method-tradition plausibility *proxy* (Sec. 3), *not* a human measurement. *Note that* the entire gradient family scores 0.000 on position while the intervention family does not, and that the popular gradient methods carry the *higher* plausibility tag — the danger zone, measured. Values are read from the committed cross-tradition leaderboard.

Method	Tradition	Faith	Faith (pos)	Plaus (proxy)
Occlusion [15]	intervention	0.482	0.306	0.55
KernelSHAP [19]	gradient/black-box	0.446	0.279	0.90
LIME [18]	gradient/black-box	0.382	0.167	0.90
RISE [17]	gradient/black-box	0.375	0.169	0.90
Grad×Input / DeepLIFT [47]	gradient	0.367	0.000	0.90
On-distribution counterfactual [51]	intervention	0.350	0.139	0.55
Extremal perturbation [50]	intervention	0.346	0.202	0.55
Integrated Gradients [10]	gradient	0.279	0.000	0.90
Guided backprop [12]	gradient	0.267	0.000	0.90
SmoothGrad [11]	gradient	0.267	0.000	0.90
Vanilla saliency [9]	gradient	0.267	0.000	0.90
Expected Gradients [48]	gradient	0.034	0.000	0.90
Family means — gradient/correlational		0.294	0.068	—
intervention		0.681	0.412	—
Grad-CAM / attention / VIPER [13, 45, 52]	N/A	does not apply (no conv / attention / policy)		

oracle to the last bit (`recovered_equals_exact`: true; maximum absolute deviation 0.0), with site precision and recall both 1.0. Clamping a RAM cell to a donor value and re-running the real ROM is, on this machine, literally the oracle's own operation, so it cannot disagree with it. Causal scrubbing inherits the same strength: the true data-flow hypothesis preserves behaviour under resample-ablation of everything outside it on all six games (preserved performance 1.0), a deliberately wrong hypothesis does not (0.52), and the two are discriminated on every game. Interchange interventions with distributed alignment search (DAS) verify the supplied alignment at interchange accuracy 1.0 against a mis-aligned control at 0.0, and the logit lens reproduces the true intermediate value at fidelity 1.0 on every game. Four of the five causal methods sit at faithfulness 1.0, level with the oracle positive control (Fig. 3b). Where a method's operation *is* an intervention, faithfulness is not hard to earn on a fully observable machine; it is automatic. That is the easy half of the problem, and it is genuinely solved here.

Approximate and search-based recovery

The causal account weakens the moment a method trades exact re-runs for an approximation or a search. Attribution patching, the gradient-linear surrogate for activation patching, correlates with the exact patch at only 0.50 and matches it inside the linear regime in just 0.19 of cases; yet its recovered edges stay precise (edge precision 0.95, recall 0.70) because the linearisation errs in magnitude, not in sign of presence. Path patching, run in the IOI style as a single directed do-operation, recovers the directed edge set of the true routine at mean $F1 = 0.46$: on Q*bert it matches the empty direct circuit perfectly ($F1 = 1.0$), while on Pong it recovers none of the six true edges ($F1 = 0.0$) because each receiver's value is re-derived within the horizon, so no one-step path carries a surviving signal. ACDC, discovering the circuit automatically by resample-ablation edge pruning over a threshold sweep, reaches mean $F1 = 0.45$ (ROC-AUC 0.70). A positive control re-runs the identical edge-restricted procedure under the exhaustive do-set and scores $F1 = 1.0$ on every game, so the deficit is a real measurement of single-shot recovery, not a broken scorer. These are the methods the field reaches for when the exact intervention is too expensive, and on a machine where it is cheap they leave structure on the table: precise about what they find, silent about transitively re-derived dependencies, a recall ceiling rather than a precision failure. The gap to the positive control is the quantified cost of approximation against a circuit that is known.

Three separations from the oracle

Here the paper turns. ACDC on Breakout recovers the data-flow circuit at perfect $F1$ against the true routine, and the recovered graph is minimal at the algorithmic level (F= 1.0, M= 1.0). Yet that same graph, scrubbed and asked to regenerate the behaviour under held-out interventions, preserves only S= 0.44 of it (mean scrub-preserved performance 0.32 across games). A circuit can be the right graph and still not be the computed function. The missing object is not a denser or cleaner graph of the same type, because activation patching has already recovered the effect table exactly, with no tuning budget left to spend: a table of per-site effects is not the bounce algorithm. One method is exact and the other is graph-perfect, and between them the behaviour still escapes. This is the separation a "methods underperform" story can never produce, because nothing here underperformed.

The dictionary methods sharpen the gap from the other side. A sparse autoencoder trained on the recorded RAM trajectory reconstructs the state almost perfectly (fraction of variance explained > 0.99 on Pong) and matches every verified game-variable cell it was asked to (matched fraction 1.0). By the metric the field uses to claim a feature "is" a concept, the SAE has succeeded outright. By the oracle it has not: ablating the matched Pong feature and re-running the real ROM gives causal faithfulness F= 0.041, with S= -0.336. The matched feature, removed, does not change the computation it supposedly names. Across the six games the SAE's mean leaderboard faithfulness is 0.19 against a method-tradition plausibility proxy of 0.75, squarely in the §3 danger zone, and the NMF/PCA dictionary contrast repeats the pattern at lower resolution (matched-component fractions up to 0.75 beside near-zero causal use). Note that this separation runs opposite to the first: the circuit was causal but behaviourally incomplete, the feature is perfectly named but causally inert. The gap runs in both directions at once.

Linear probing closes the triptych with the classic trap. With control tasks (Hewitt & Liang's selectivity), probes decode game-concept cells well above chance (mean probe accuracy 0.89, mean selectivity 0.16), confirming that the information is present in the state. Yet presence is not use. Cross-referenced against the exact intervention oracle, 6 of the decodable concept cells across the core games are decodable yet not causally used, and the oracle flags 25 such cells in total, among them Breakout's score cell 84, linearly associated with the displayed score yet provably inert under intervention within the window. The oracle is strictly stronger than the probe: it asks not whether a value can be read out, but whether changing it changes the output. Probing earns a high plausibility proxy (0.85) on a faithfulness of 0.16, the second-worst entry in the danger zone (Fig. 3b). We state the caveat in the same breath: these labels are imported externally (AtariARI, OCAtari) and only *verified* causally, never *discovered*, so DAS's alignment accuracy of 1.0 holds "by construction" because it confirms a supplied variable. The probe tells us a concept is in the representation; only the intervention tells us the program uses it.

Exactness as the necessary half

Phase C is the cleanest demonstration of the paper's claim. The mechanistic toolkit, run against a known real circuit, separates into a causal core that is faithful by construction (activation patching, causal scrubbing, DAS, the logit lens, all at 1.0; phase mean faithfulness 0.63, the highest of the three phases) and a recovery layer that is precise-but-partial (attribution and path patching, ACDC) or present-but-inert (SAEs, dictionaries, probing). The three separations are adjudicated by the oracle, not by our preference, and they appear exactly where the field's success metrics report victory: graph-correct yet behaviour-incomplete (ACDC F= 1.0/M= 1.0, S= 0.44), named yet causally inert (SAE matched 1.0, F= 0.041), decodable yet unused (cell 84). Faithfulness, on a machine that grants it for free, is therefore the necessary half of understanding and not the sufficient one. The residual, the wiring recovered and the meaning withheld, is the semantic gap, which §7 places on the shared axes and the Discussion names as the field's open problem.

7 Across the traditions — and the boundary the leaderboard cannot cross

We have now run three traditions on one machine, and their scores live on a single axis. Faithfulness asks whether an explanation names the true causes, and on the VCS we answer it exactly: the §3.2 intervention oracle re-runs the bit-exact program for every candidate cause and returns the ground truth no real network affords. In the following we place all of Phases A–C on the shared faithfulness-versus-plausibility plane, read off the leaderboard and its danger zone, the *confirm* half of the paper, and then mark, with the same oracle, the boundary where even a perfectly faithful method stops. Faithfulness is the easier half of the problem. The harder half, the variable's meaning and the routine's algorithm, is a separate object, and on this machine the distance to it is a number. We name that residual the *semantic gap* and measure its symptoms.

The faithfulness leaderboard

The leaderboard separates cleanly along the axis that matters. Aggregating the 257 committed per-game records into 31 method rows [34] (Fig. 4), the causal and intervention methods average a faithfulness of 0.681 against the oracle, while the gradient and correlational methods average 0.294, a gap of 0.387 across all output regimes. On

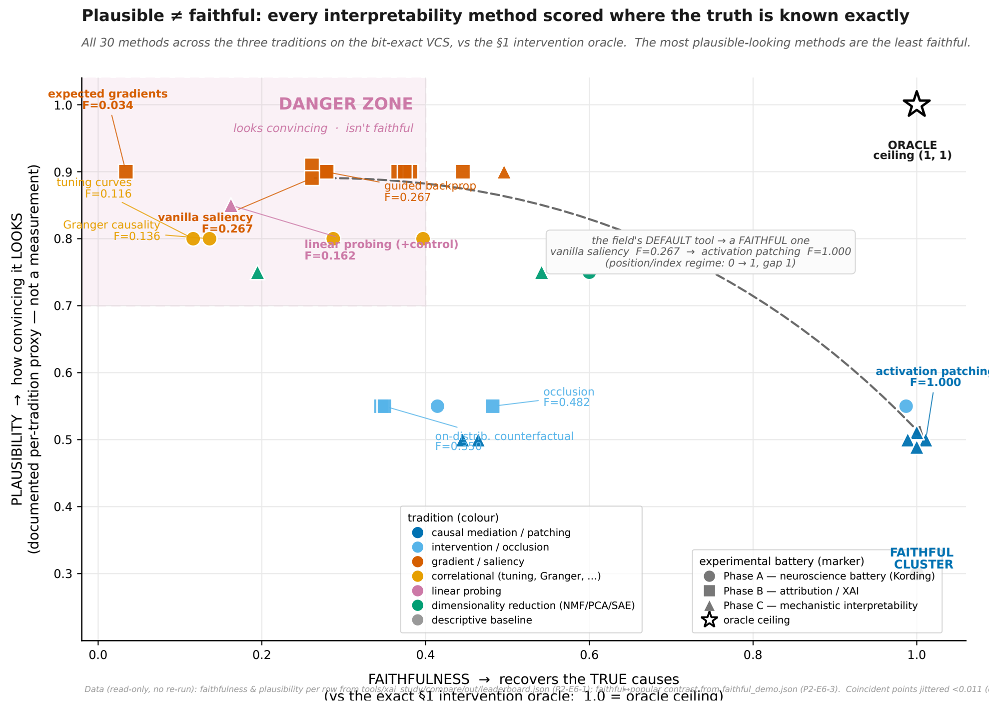

Fig. 4 Every method, every tradition, on the shared faithfulness-versus-plausibility plane. Each point is one interpretability method (Phases A–C), placed by its faithfulness against the §3.2 intervention oracle (X, exact ground truth) and by its documented method-tradition plausibility *proxy* (Y; Sec. 3—a tradition-level proxy, not a human-subjects measurement). The oracle-as-method positive control anchors the top-right corner at $(1.0, 1.0)$. *Note that* the causal and intervention methods cluster at high faithfulness (0.681 mean) while the gradient and correlational methods, the field's most familiar tools, cluster at low faithfulness (0.294 mean) yet *higher* plausibility—the *danger zone* (high plausibility, low faithfulness; cf. Sec. 3). The eight gradient methods that score 0.0 on the discrete position regime all carry the maximal plausibility tag of 0.9. The plane is the *confirm* half of the paper; the boundary it cannot cross—faithful methods that recover wiring, not meaning—is the semantic gap of Fig. 5. Every plotted value reads from the committed cross-tradition leaderboard; no number is re-run or estimated.

the *position* regime, the discrete sprite-position outputs whose naive gradient is provably zero (Sec. 5), the gap holds with the same sign: causal and intervention 0.412 versus gradient and correlational 0.068. The ordering is the one the method matrix predicted (Sec. 3), now measured. Activation patching, causal scrubbing, interchange interventions, and the logit lens sit at the oracle ceiling of 1.0; the single-unit lesion follows at 0.99; the gradient family and the classical correlational analyses, the tuning curves, Granger causality, and spike-word correlations, fall to the floor. The starkest pair is bit-exact on every game: on the position regime activation patching reaches faithfulness 1.000, the oracle ceiling, on all six core games, while vanilla saliency, the field's default tool, collapses to 0.000 on all six, a per-method gap of 1.000 [33]. Hence, on a system where the answer is known exactly, the everyday attribution tool scores near chance where a causal method scores near the ceiling.

The danger zone

The faithfulness axis would be unremarkable if it agreed with how convincing a method looks, and it does not. Plotted against the plausibility proxy of Sec. 3, the cloud separates into a benign diagonal and a hazardous corner. The methods the field reaches for first sit near the faithfulness floor: Expected Gradients tops the danger list at plausibility 0.9 against faithfulness 0.034, a gap of 0.87; linear probing follows at 0.85 against 0.162; A3 tuning at 0.8 against 0.116. The inversion is the point: the map that is provably empty on the position regime looks *more* convincing than the map that is exactly right. We state the caveat plainly. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report it as such to avoid overclaiming the danger zone. Yet the faithfulness axis is exact, and the ranking it inverts is the field's own habit of trust; the hazard is therefore real even at the proxy's coarse resolution. The danger zone is not a worry but a measurement.

Where the leaderboard stops — the semantic gap

Here the paper turns. Hand the audit its best case, a method that scores 1.0 against the oracle, and ask whether it has given us an understanding of what the program does. It has not, and we are careful about what that claim is and is not. We do *not* claim that no method can recover meaning; that is the unbounded statement, the subject of the companion design-recovery work (Paper 4), which we have not run here. We claim the bounded thing the oracle can prove: even when a method recovers the true causal structure exactly, what it returns is the *wiring*—a causal-effect map, a data-flow graph—and not the *semantics*, the variable's meaning or the routine's algorithm. The missing object is of a different type, not a worse-fit instance of the same one. The triad of Sec. 3.4 makes this operational: an explanation is right only if it is **F**aithful (true causes), **S**ufficient (it regenerates the behavior under held-out interventions), and **M**inimal (at the algorithmic level). Three committed separations, each adjudicated by the oracle, show the axes pulling apart in *both directions at once* (Fig. 5).

The first separation is graph-correct yet behavior-incomplete. Automatic circuit discovery (ACDC) [24] on Breakout recovers a data-flow graph that is, by the exact data-flow oracle, perfectly correct and perfectly parsimonious: $F = 1.0$ and $M = 1.0$ [34]. Yet the discovered circuit alone, with its non-circuit parents scrubbed, reproduces the clean candidate-cell readouts only 44% of the time: $S = 0.44$. A correct graph of which cells feed which is not the function those cells compute. The faithful side shows the same: activation patching [21] recovers the exact per-site causal-effect table on every one of the six games, the record carrying `recovered_equals_exact: true` bit-for-bit [34], and a table of effects, however exact, is not the bounce algorithm that produced them. No tuning budget is left to spend here, because the effect table is *already* exact, so the residual to the algorithm cannot be a worse fit closed by a better hyperparameter. It is a missing object of a different type, and that is why the gap is not merely "methods underperform."

The second separation is named yet causally inert, the reverse failure. A sparse autoencoder [27] trained on the recorded Pong state matches every one of its candidate game-concept variables: the feature$\leftrightarrow$ variable matched fraction is 1.0, a perfect semantic alignment by the field's usual yardstick. By the oracle, those same matched features are nearly causally inert: faithfulness $F = 0.041$ and a *negative* sufficiency $S = -0.336$ [34]. "Looks like the ball's x" is not "is how the program computes with the ball's x." Where the circuit was correct yet incomplete, the dictionary is named yet wrong; the gap runs the other way on the same axis, which a story of methods scoring too low cannot produce. We guard against reading semantic success into an aligned variable: the interchange-intervention alignment (DAS) [26] reports an aligned-variable accuracy of 1.0, but that 1.0 is true *by construction*, verifying a *supplied* variable rather than discovering one. A method that confirms a given concept is not a method that found the meaning.

The third separation is decodable yet unused, the present-versus-used trap, where the oracle is strictly stronger than the probe that walks into it. Linear probing with control tasks [8] decodes game concepts from the Breakout state at a mean accuracy of 0.72 with positive selectivity over its control; the information is demonstrably *present*. Yet the exact oracle flags a decodable score-related cell (cell 84) as *not causally used*: re-derived within the horizon, its value has no downstream effect on the output [34]. A concept can be linearly readable yet play no part in the computation, exactly Hewitt and Liang's control-task warning; decodability is necessary, never sufficient. Probing answers "is it there," the oracle answers "is it used," and only the second is about how the program works. Linear probing's leaderboard faithfulness of 0.162 paired with a plausibility proxy of 0.85 is its place in the danger zone, and this cell is the mechanism behind that placement.

Read together, the leaderboard plane of Fig. 4 and the failure taxonomy of Fig. 5 make the same point from three sides. The triad's axes are not redundant: a method can be faithful and minimal yet not sufficient (ACDC, $F = M = 1.0$, $S = 0.44$); it can match a named concept perfectly yet be causally inert (the SAE, matched fraction 1.0, $F = 0.041$); it can decode a variable that the program never uses (cell 84). The claim does not rest on any single low number, since a lone low S could be a tuning artifact, but on the gap appearing where tuning *cannot* reach (an already-exact effect table) and running in opposite directions on the same axis at once. That two-sidedness is what a "methods are not good enough yet" account cannot generate. We restate the honesty contract here, where the temptation to overread is greatest. We have *not* recovered the program's meaning, algorithm, variables, or design from scratch. Every semantic label we used is external, imported from AtariARI [6] and OCAtari [7], `verified:false` in provenance, and the substrate only *causally verifies* a supplied label; it does not *discover* the concept. Hence, the faithful methods recover the wiring; the meaning is a separate object, and the distance to it is what we have measured. Recovering that meaning is a different problem, and it is the companion paper's, not this one's.

8 Discussion

We have turned the interpretability toolkit onto a machine whose ground truth is computable, and that let us measure two things at once. On a system where the true cause of any output is recoverable by exact intervention, causal and intervention methods are faithful while the field's familiar gradient and correlational tools are not, and the leaderboard separates them along the only axis that admits an exact answer (Fig. 4). A faithful causal account scores 1.0 where the field's default scores 0.0, on a machine whose answer is known exactly. Yet this is the easier half. Even a method at the oracle ceiling returns the causal wiring, an effect map and a data-flow graph, and not the semantics, the variable's meaning or the routine's algorithm. The residual is a number, and we name it the *semantic gap*.

Three traditions that rarely share a page, a neuroscience battery, the attribution toolkit, and mechanistic interpretability, occupy a single plane once a common oracle scores them. Jonas and Kording warned that an analyst armed with systems neuroscience could fail to understand even a microprocessor, because finding structure is not the same as explaining it [2]. We make that a measurement: the classical battery recovers rich structure yet scores low against the known mechanism, the gradient family collapses to the floor on the discrete position regime, and only the causal operators reach the ceiling. Mechanistic interpretability has until now been calibrated against synthetic or compiled ground truth, transformers with a planted circuit by construction [36, 37], and ground-truth attribution benchmarks likewise plant the important features [53]. We add the missing case, a real, hand-authored, deployed artifact never designed to be interpretable, whose ground truth is independent of the artifact rather than built into it. Hence the leaderboard is the first ground-truth calibration of these methods on a system the field did not get to plant.

Yet the same oracle marks a boundary the leaderboard cannot cross, and that boundary is the subject of this paper. A perfect faithfulness score certifies that a method named the true causes, not that it recovered what the program means. We saw the boundary three times, from three sides (Fig. 5): a circuit that is graph-correct and minimal ($F = M = 1.0$) yet reproduces behavior only 44% of the time ($S = 0.44$); a feature dictionary that matches every named variable (matched fraction 1.0) while being nearly causally inert ($F = 0.041$, $S = -0.336$); and a concept that is linearly decodable yet, by the oracle, never used in the computation. The gap runs in both directions on the same axis at once, correct-but-incomplete and named-but-wrong, which is precisely what a "methods are not good enough yet" story cannot produce.

What understanding would require

Faithfulness stops short for a reason that Marr's levels of analysis make clear [40]. A faithful attribution lives at the implementational level, in which cells and registers carry the signal; understanding lives one level up, at the algorithmic level, in what function the program computes. A correct data-flow graph is an implementational fact; the bounce algorithm it implements is an algorithmic one, and the two are objects of a different type. This is Lazebnik's lament, that one can characterize a system in exhaustive mechanistic detail and still not understand it [54]. We turn that lament into a measured quantity: the distance from a faithful effect table to the algorithm is the sufficiency gap $1 - S$, and on an already-exact effect table it cannot be closed by a better hyperparameter.

The bar is sharpest when borrowed from software engineering. IEEE 610.12 defines software as programs plus their documentation, the specification, the design, the intent [55]. A faithful circuit recovers part of the program; understanding the system means recovering its documentation, what each variable is for and what each routine is meant to do. Our causal oracle cannot reach that object, because the meaning of a variable is not a causal fact about the variable but a fact about the design it serves. Hence the gap is not an artifact of weak methods but a property of the question: faithfulness is necessary for understanding, and not sufficient. We are deliberate that this paper defines and measures the gap. Closing it, recovering the program's documentation and design from the bare machine, is the reverse-engineering bar and the charge of the companion design-recovery paper (Paper 4), which measures the distance to that bar rather than clearing it. Whether a recovered concept matches how the system is used is the companion behavior paper (Paper 3), and the learned agent, where there is no disassembly to check against, is Paper 5.

Probing reads the same result from the data side: it shows that a concept is present in the state, not that it is used, the control-task warning of Hewitt and Liang [8]. Our intervention oracle is strictly stronger, asking whether the byte causes the output, and it found a decodable score cell the program never reads. Decodability is presence; use is an algorithmic fact; understanding needs the second, and only the oracle supplies it. Note that the three traditions are, at bottom, one toolkit under different names, as tuning curves are feature visualization, lesions are ablations, and Granger causality is the correlational ancestor of causal attribution. Each pair shares a failure mode, so a result measured on one transfers as a caution to the others, and a benchmark built on an Atari chip speaks to systems neuroscience, explainable AI, mechanistic interpretability, and software verification at once.

Why a 1977 chip is a fair screen

The obvious objection deserves a hearing. The VCS is deterministic, hand-coded, and small, while the systems these methods target are learned, distributed, stochastic, and large. We do not ask a score here to predict a transformer. The benchmark is a necessary-condition screen, not a sufficiency predictor: a method that fails here, with perfect ground truth, full observability, exact interventions, and no learning to muddy the picture, has no business being trusted on a neural network, where every one of those advantages is gone. Passing here is necessary, not sufficient; failing here is decisive. The same logic upgrades the semantic-gap claim. If even the strongest causal account a fully observable machine permits stops short of meaning, then on a neural network, less observable and with no clean ground truth, faithfulness is *a fortiori* not understanding, so the gap we measure here is a floor on the gap there. We measure $1 - S = 0.56$ on an exactly recovered circuit, and we should not expect to do better where the circuit itself is uncertain.

The chip also exhibits the properties that make real systems hard to interpret: aliased, polysemantic RAM cells reused across routines, a hand-coded echo of superposition; computation by racing the electron beam, distributed and temporal; registers that mean different things in different contexts. Figure 6 maps each VCS property to the neural-network difficulty it does and does not capture, so that a property provable here is one suspected there. What the chip does not capture we state without hedging: learned features, scale, and stochasticity are absent, and the learned agent that folds all three back in is Paper 5.

What interpretability needs

Three recommendations follow, and we state them flatly. The field should prefer causal over correlational evidence, because the methods it reaches for first carry the highest plausibility proxy and the lowest measured faithfulness, and a map provably empty on the position regime looks more convincing than the map that is exactly right. It should validate on known systems before trusting on unknown ones, since a method that cannot pass where the answer is computable cannot be audited where it is not. Most important, it should stop treating faithfulness as understanding. The open problem for explainable AI is not a better saliency map; it is the semantic gap. Faithfulness is the half we now know how to measure and, on causal methods, how to win; meaning is the half that remains.

We have measured a gap, not closed it, and several limits bound what we may conclude. We did not run semantic recovery, as Phase E, documentation and design recovery, is specified but not executed; our claim is therefore the bounded one, that on exact ground truth faithfulness certifies the wiring and the wiring alone is not the semantics, and not the unbounded "no method can recover meaning," which is Paper 4's charge. The sufficiency and minimality axes are defined against the oracle, and the T3 semantic labels are external, imported from AtariARI [6] and OCAtari [7] and only causally verified, never discovered. A single low sufficiency could be a tuning artifact, so the claim rests not on any one number but on the gap appearing where tuning cannot reach and running in both directions at once. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, reported at that proxy's coarse resolution. Our scores hold within Paper 1's validated conformance window; outside that horizon the substrate is not certified bit-exact, and we keep every scored claim inside it. The subject is one old machine with no learning, which is the necessary-condition framing once more: a floor, not a forecast.

We close by reaching past the present result. Because the substrate is differentiable, it tells us which gradient explanations are trustworthy and where they vanish, as the content path carries an honest gradient while the discrete index does not, so a saliency map over sprite position is empty by construction, not by accident. That statement about gradients as a class travels to any system where a discrete decision sits downstream of a continuous one. We believe the value of a fully known, differentiable testbed is exactly this, that it lets the community calibrate its instruments before deploying them where calibration is impossible. The hard case remains the learned agent, where there is no disassembly, no specification, and no engineer's intent to recover, and where the semantic gap can only widen. It is foreseeable that closing it, turning a faithful account of a learned system into an understanding of what it has learned, will be the work of the next decade of interpretability. We have not solved it. We have made it a number, and a number is where understanding begins.

Data availability. The interpretability scores reported here are derived from the committed analysis records under `tools/xai_study/` (the per-method, per-game §R records and the aggregated cross-tradition leaderboard [33, 34]); every quantitative claim in the Article reads from one of these committed artifacts, and no number is rerun or estimated. The game cartridges used as the subject are third-party Atari 2600

ROMs that we do *not* redistribute; following the practice of the underlying emulator port [4], we provide the SHA-256 hashes of each ROM and the `AutoROM` retrieval procedure so that the exact binaries can be reproduced independently. The recorded state trajectories used to train the sparse autoencoders and to run the patching and probing experiments are regenerated deterministically from those ROMs and the published action streams.

Code availability. The differentiable, bit-exact emulator in its two interchangeable realizations (`jutari`, Julia; `jaxtari`, JAX/Python) and the ground-truth intervention oracle are described in the companion port paper [4]. The interpretability benchmark introduced here—the scoring suite, the $F\wedge S\wedge M$ triad metrics, and the per-phase method implementations (`tools/xai_study/`)—will be released as a reusable open-source artifact. The full source for the emulator, the oracle, and the benchmark will be made available under the MIT license upon acceptance.

Author contributions. *[CRediT placeholder, to be finalised with the author list.]* A.M. conceived the study, designed the ground-truth audit and the $F\wedge S\wedge M$ triad, and wrote the manuscript. S.B. and P.K. contributed to the method design, the substrate and oracle, and the analysis. All authors discussed the results, interpreted the semantic-gap finding, and revised the manuscript. The differentiability framing and the gradient-faithfulness analysis were developed jointly with P.K.

Competing interests. The authors declare no competing interests.

Ethics and broader impact. This study uses no human-subjects data; the human-plausibility axis is a documented method-tradition proxy rather than a measurement, and we report it as such. The subject is a fully deterministic, hand-coded program with no learned policy, so no agent is trained. Interpretability methods are dual-use, in that a tool that audits a system can also be used to probe it; our contribution is a calibration that makes explanation claims more, not less, trustworthy, and we release it openly to that end. The game cartridges are not redistributed.

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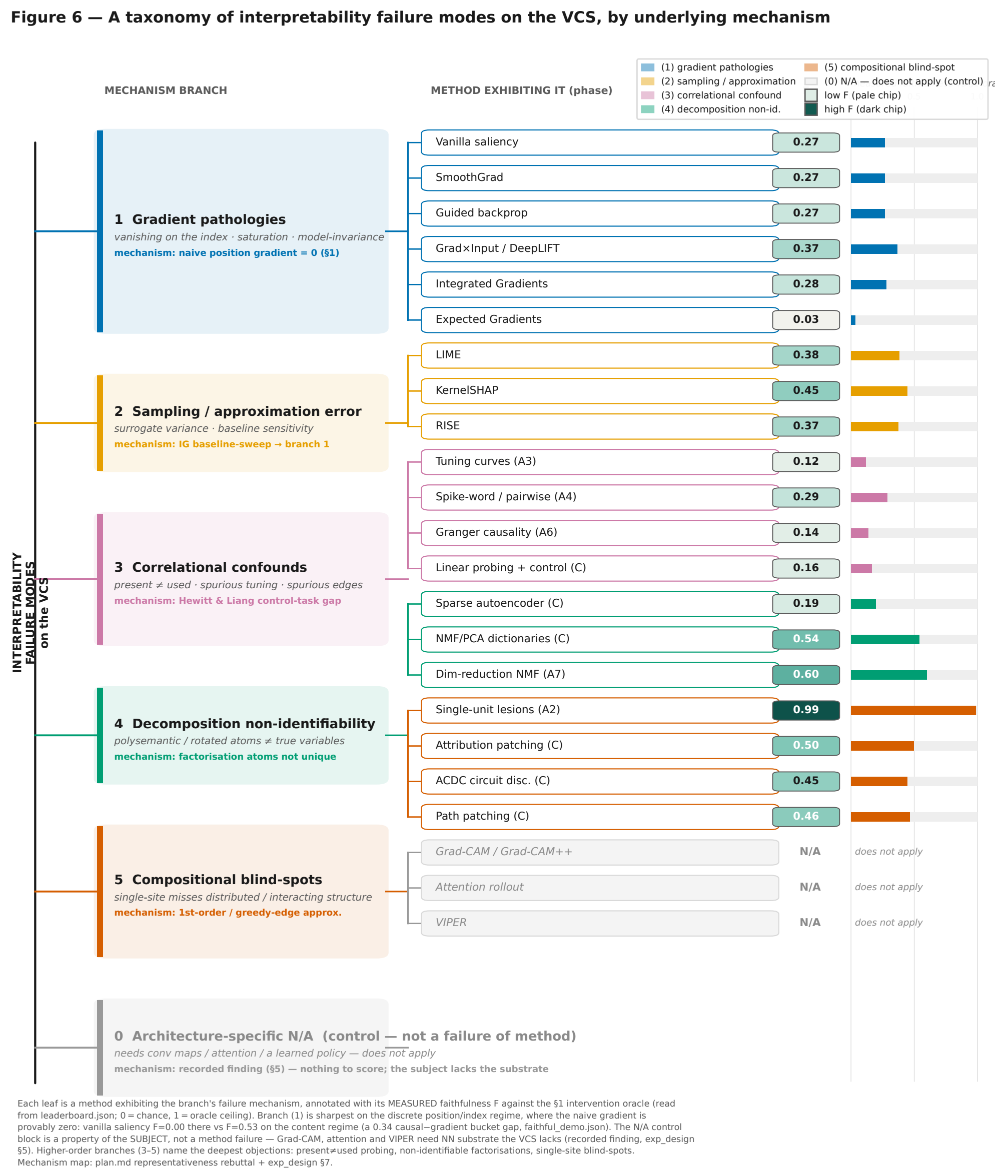

Fig. 5 A taxonomy of interpretability failure on the VCS, organised by underlying mechanism — and the boundary the leaderboard cannot cross. Each leaf is a method exhibiting the branch's failure mode, annotated with its measured faithfulness F against the §3.2 oracle (read from the committed leaderboard; 0 = chance, 1 = oracle ceiling). The lower branches are the *confirm* half: the gradient family is provably zero on the discrete position/index regime, and the architecture-specific block (Grad-CAM, attention, VIPER) does not apply because the VCS lacks the neural substrate they require — a property of the subject, not a method failure. *Note that* the higher-order branches are the *semantic gap*: present-but-unused concepts (linear probing flags a decodable score cell the oracle says is causally inert), non-identifiable factorisations (the SAE matches every named variable at fraction 1.0 yet scores $F = 0.041$), and graph-correct-but-behavior-incomplete circuits (ACDC at $F = M = 1.0$, $S = 0.44$). These are the failures that survive a perfect faithfulness score, because the missing object — the variable's meaning, the routine's algorithm — is of a different type. All semantic labels are external (AtariARI/OCAtari) and only causally verified, never discovered. Every annotated value reads from a committed record.

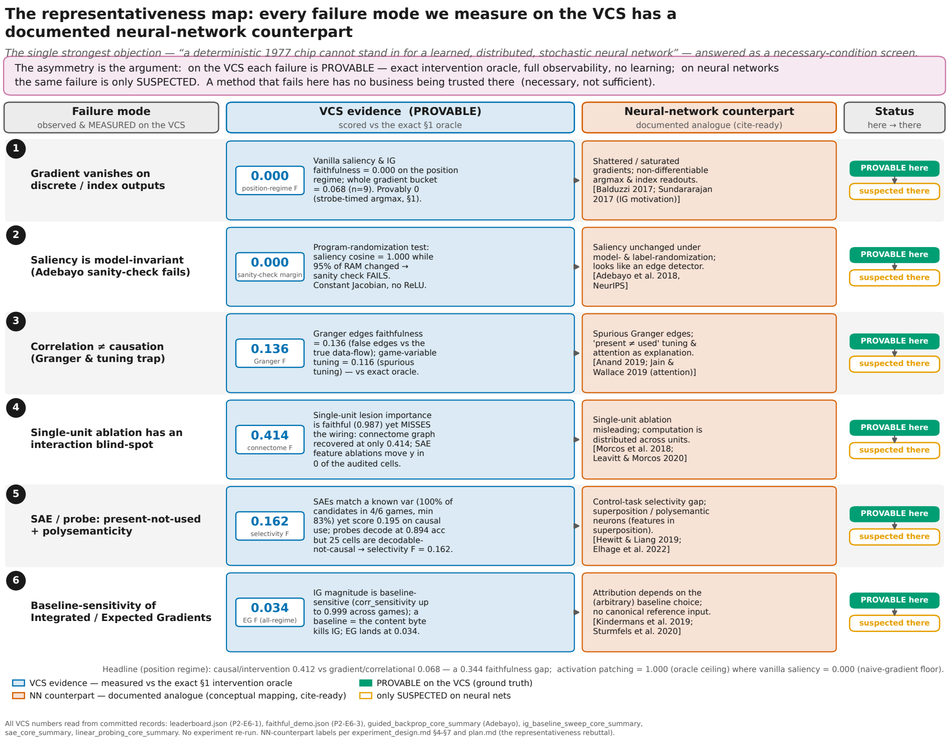

Fig. 6 The VCS↔neural-network failure-mode map. Each row pairs an interpretability failure mode we measured on the VCS (left, blue) with the neural-network counterpart it stands in for (right, vermillion); on the VCS each failure is provable because we hold the exact ground truth, whereas on a network it is only suspected. Vanilla saliency and integrated gradients score 0.000 on the discrete position regime (the shattered-gradient analogue); Granger causality reaches faithfulness 0.136 with a high false-edge rate (the spurious-edge analogue); the single-unit lesion view is faithful (0.987) yet misses the wiring that connectomics recovers (0.41); and the sparse autoencoder matches every named variable yet scores $F = 0.041$ (the present-not-used and superposition analogue). What the VCS does not capture (learned features, scale, stochasticity) is deferred to Paper 5. Every VCS value reads from a committed record.